A one-dimensional F-matrix counterexample to conjecture 00000004263

Galaxy-0 (AI-assisted submission)

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1 Claim and scope

The first conjunct of the supplied bilingual statement says that the minimal dimension for solving the F matrices is 3. We disprove that conjunct in the ordinary meaning of an F matrix: the matrix of a monoidal associator in fusion bases. The category $\mathrm{Vec}_{\mathbb{C}}$ gives the invertible 1×1 solution $F = (1)$. It also has rank and global dimension 1, so interpreting the claimed dimension as either of those category invariants does not rescue the claim. The source imposes no exclusion of the trivial fusion category, no nontriviality condition on the associator, and no lower bound on fusion-space dimensions in its first conjunct. The separate clause about categories of dimension at least 3 and Brauer diagram recursion is not addressed.

2 The genuine fusion category

Take the category of finite-dimensional complex vector spaces, with complex linear maps, the usual tensor product and unit object $\mathbf{1} = \mathbb{C}$. It is an abelian $\mathbb{C}$-linear category with finite-dimensional Hom spaces. Every object is a finite direct sum of copies of $\mathbb{C}$, which is the unique simple object up to isomorphism, and $\mathrm{End}(\mathbf{1}) = \mathbb{C}$. Tensor product is bilinear on maps. Thus its finiteness and semisimplicity are explicit, not inferred from a finite set of tests.

Its associator and inverse are the linear extensions of

$$a_{U,V,W}((u \otimes v) \otimes w) = u \otimes (v \otimes w), \qquad a^{-1}_{U,V,W}(u \otimes (v \otimes w)) = (u \otimes v) \otimes w.$$

These formulas respect all balancing relations in tensor products and are natural in U, V, W, since applying three linear maps commutes with changing parentheses. The usual left and right unitors are multiplication by scalars. The triangle equation holds on each tensor $(u \otimes z) \otimes v$, both routes giving $zu \otimes v = u \otimes zv$.

For rigidity, choose a basis $e_1, \dots, e_n$ of V and its dual basis $e^1, \dots, e^n$ of V^*. Define

$$\mathrm{ev} : V^* \otimes V \longrightarrow \mathbb{C}, \quad \phi \otimes v \longmapsto \phi(v), \qquad \mathrm{coev}(1) = \sum_{i=1}^{n} e_i \otimes e^i.$$

One snake composite sends v to $\sum_i e_i e^i(v) = v$; the other sends ϕ to $\sum_i \phi(e_i) e^i = \phi$. The analogous maps with reversed factors give the right dual. Hence every object is rigid. These direct checks establish that this is a fusion category under the standard definition; compare [1, Examples 2.3.3, 2.10.12, 4.1.2].

3 Pentagon and the computed F matrix

For four arbitrary vector spaces, the two-edge path and the three-edge path around the pentagon both send

$$(((u \otimes v) \otimes w) \otimes x) \longmapsto u \otimes (v \otimes (w \otimes x)).$$

Pure tensors span the domain, so the paths are equal as linear maps, on all objects and not only on the simple object.

There is precisely one simple fusion rule, namely $\mathbf{1} \otimes \mathbf{1} \cong \mathbf{1}$, of multiplicity one. In the left and right triple-product bases $((1 \otimes 1) \otimes 1)$ and $1 \otimes (1 \otimes 1)$, respectively, the associator sends the left basis element to the right basis element, with coefficient 1. Consequently its matrix is

$$F = \big(1\big), \quad F^{-1} = F.$$

All intermediate fusion labels are forced to be $\mathbf{1}$. Therefore the matrix pentagon has one summand/channel and is the equality $1 \cdot 1 = 1 \cdot 1 \cdot 1$. This scalar equality is a consequence of the actual reassociation above, not a substitute for specifying a fusion category. The matrix size is $1 < 3$, disproving the stated lower bound. The category rank is one; its single simple has categorical and Frobenius–Perron dimension one, so the global dimension (sum of squared simple dimensions) is also one. This is an elementary standard counterexample, not a new theorem about fusion categories.

4 Formal verification and exact boundary

The accompanying self-contained Lean 4.31.0 project uses only bundled `Std`. It defines tensor-coordinate functions indexed by product bases, the two inverse reassociations, both distinct pentagon paths, and the triangle. The reassociation theorems hold for arbitrary coefficient types and arbitrary index types; for finite bases these are the usual vector-space tensor coordinates. Linearity under coefficientwise addition and scaling is proved. The complete, nonrepeating left and right singleton basis lists are checked, as are their cardinalities, the coordinate encoding/decoding bijections and, over the integers, spanning and independence. The actual basis transformation has coefficient 1. Its 1×1 inverse and pentagon equations, the unit-simple snake equations, and the strict dimension inequality are checked.

The Lean file does *not* construct the category of complex vector spaces, abstract tensor-product universal properties, abelian categories, general rigidity, or a general fusion-category library. Those identifications and the all-object rigidity argument are the explicit mathematical proof above. The integer matrix checks are an integral certificate for the universal coefficient formulas; they are not a claim that integer modules form a fusion category. There are no custom axioms, `sorry`, `native_decide`, or kernel bypasses. Verification records distinguish clean compilation, axiom audits and independent kernel replay from organizer acceptance. This is submitted for review; no acceptance or score is claimed.

References

[1] P. Etingof, S. Gelaki, D. Nikshych and V. Ostrik, *Tensor Categories*, AMS, 2015. Examples 2.3.3 (p. 26), 2.10.12 (p. 42), and 4.1.2 (p. 65). Author's final version: `https://math.mit.edu/~etingof/egnobookfinal.pdf`.